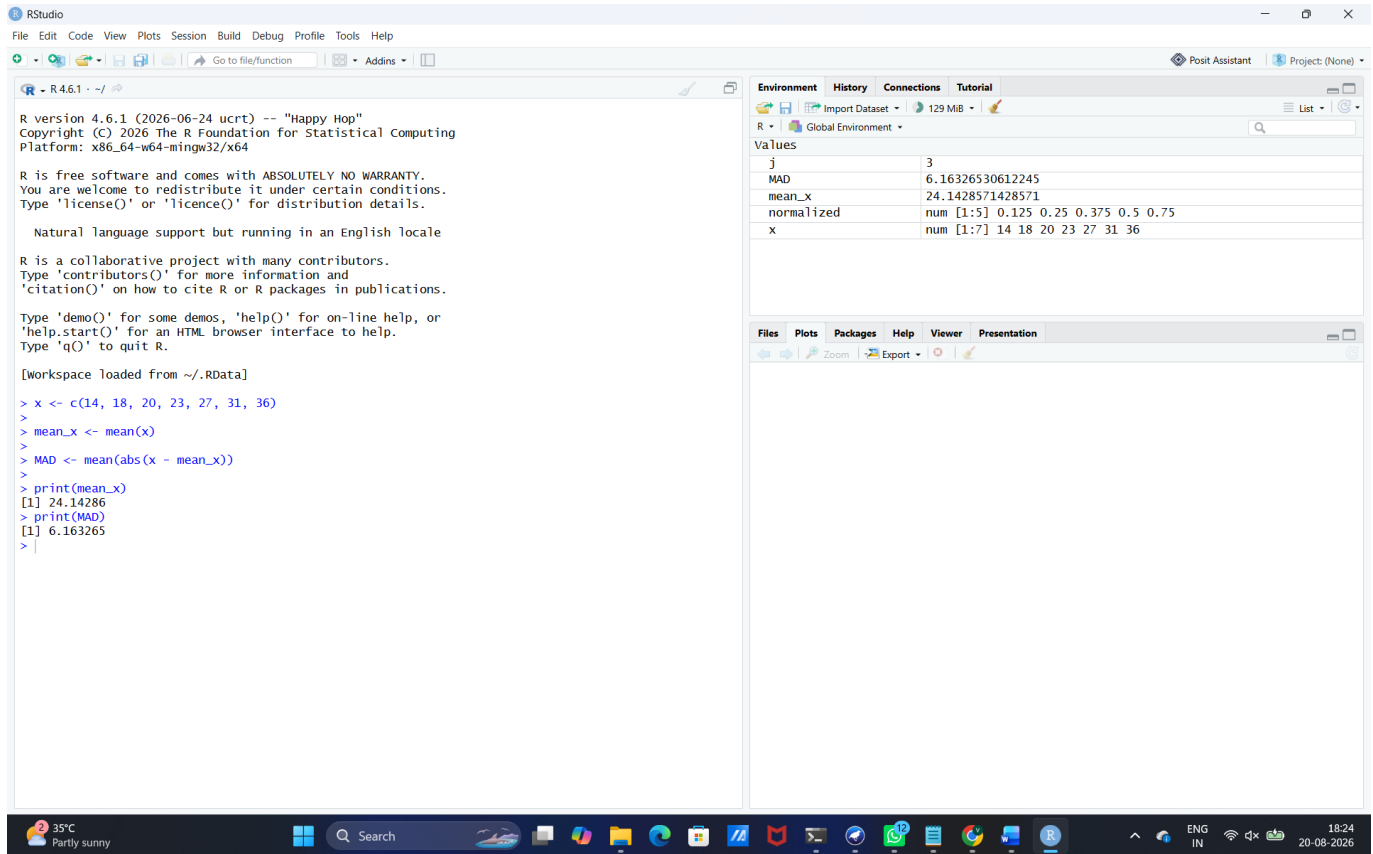


SET 4

1. R: Calculate mean absolute deviation from the mean for 14,18,20,23,27,31,36.



The screenshot shows the RStudio interface with the following content:

Console:

```
R version 4.6.1 (2026-06-24 ucrt) -- "Happy Hop"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

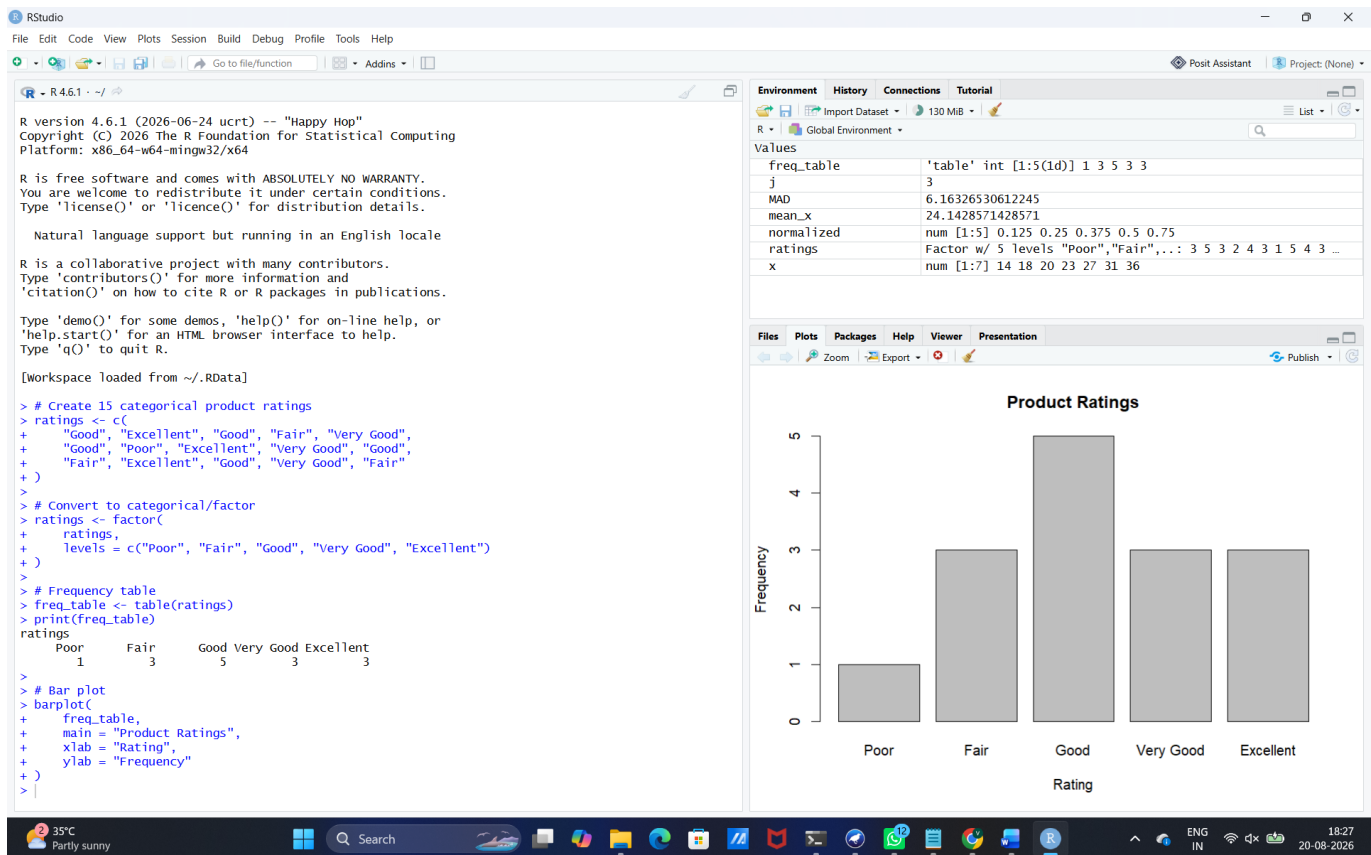
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]
> x <- c(14, 18, 20, 23, 27, 31, 36)
>
> mean_x <- mean(x)
>
> MAD <- mean(abs(x - mean_x))
>
> print(mean_x)
[1] 24.14286
> print(MAD)
[1] 6.163265
>
```

Environment:

Variable	Value
j	3
MAD	6.16326530612245
mean_x	24.1428571428571
normalized	num [1:5] 0.125 0.25 0.375 0.5 0.75
x	num [1:7] 14 18 20 23 27 31 36

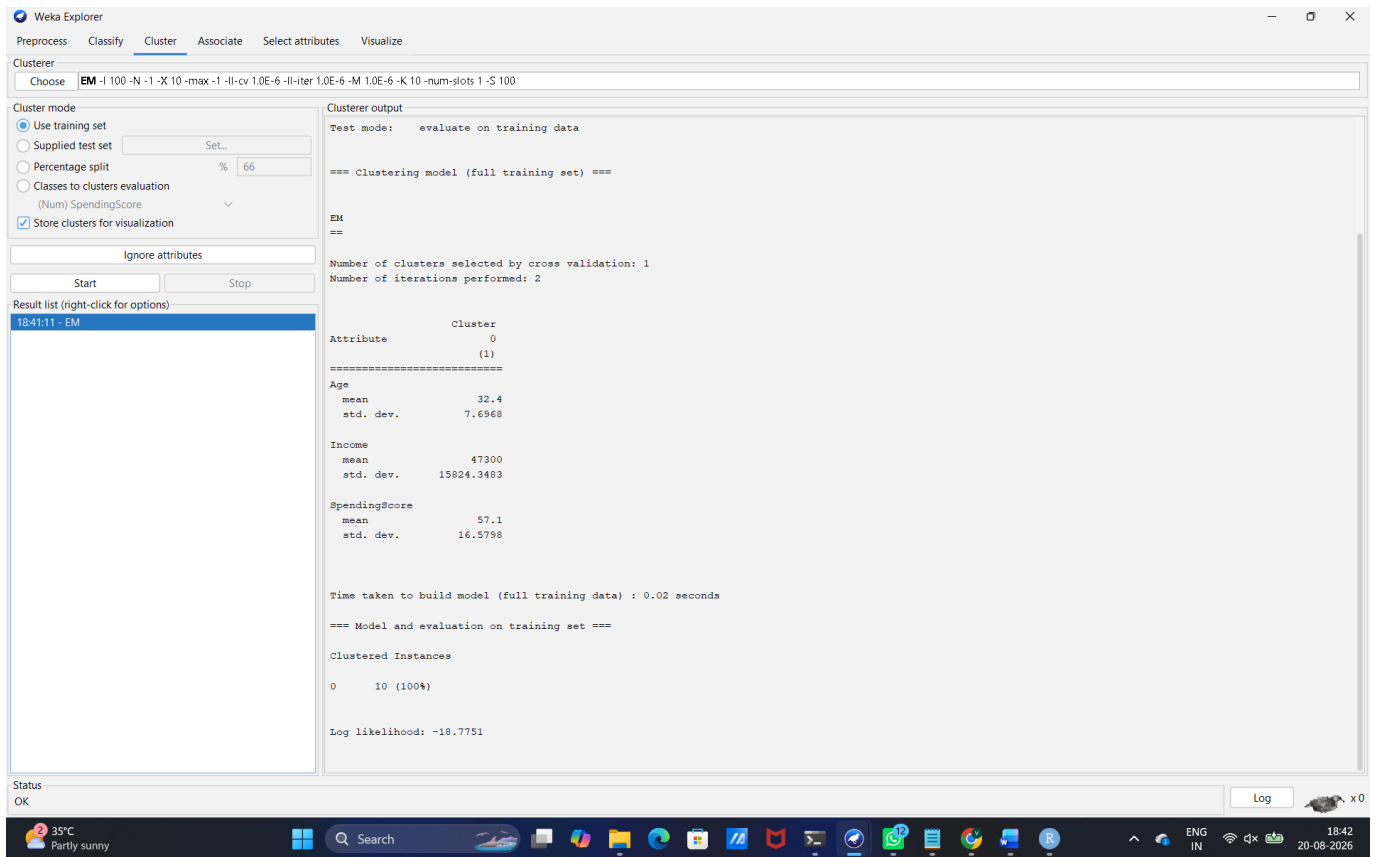
2. R: Create 15 categorical product ratings; generate a frequency table and bar plot.



3. WEKA: Apply EM clustering to a suitable numeric dataset and compare its clusters with SimpleKMeans.

Dataset: EM Clustering

```
@relation customer_em
@attribute Age numeric
@attribute Income numeric
@attribute SpendingScore numeric
@data
22,28000,35
25,32000,40
27,35000,45
30,42000,55
34,50000,60
38,58000,70
42,68000,78
46,76000,85
24,30000,38
36,54000,65
```



4. WEKA: Apply COBWEB to the same dataset and inspect the hierarchical cluster structure.

Dataset: COBWEB

```
@relation customer_hierarchical
@attribute Region {North,South,East,West}
@attribute Visits numeric
@attribute Purchases numeric
@attribute Rating numeric
@data
North,12,3,3
North,18,5,4
South,9,2,2
South,14,4,3
East,22,7,5
East,20,6,4
West,8,1,2
West,11,3,3
```

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Clusterer

ChooseCobweb -A 1.0 -C 0.0028209479177387815 -S 42

Cluster mode

☒ Use training set

☐ Supplied test set

Set...

☐ Percentage split

%66

☐ Classes to clusters evaluation

(Num) Rating

☒ Store clusters for visualization

Ignore attributes

StartStop

Result list (right-click for options)

18:47:02 - Cobweb

Clusterer output

node 0 [8]

| node 4 [2]

| | leaf 5 [1]

| node 4 [2]

| | leaf 6 [1]

node 0 [8]

| node 7 [2]

| | leaf 8 [1]

| node 7 [2]

| | leaf 9 [1]

node 0 [8]

| node 10 [2]

| | leaf 11 [1]

| node 10 [2]

| | leaf 12 [1]

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

2

1 (13%)

3

1 (13%)

5

1 (13%)

6

1 (13%)

8

1 (13%)

9

1 (13%)

11

1 (13%)

12

1 (13%)

Status

OK

Log

 x 0